

A MODEL OF DOUBLE DESCENT FOR HIGH-DIMENSIONAL LOGISTIC REGRESSION

Zeyu Deng^{*} Abba Kammoun[†] Christos Thrampoulidis^{*}

^{*}University of California, Santa Barbara [†]King Abdullah University of Science and Technology

ABSTRACT

We consider a model for logistic regression where only a subset of features of size p is used for training a linear classifier over n training samples. The classifier is obtained by running gradient-descent (GD) on the logistic-loss. For this model, we investigate the dependence of the classification error on the overparameterization ratio $\kappa = p/n$. First, building on known deterministic results on convergence properties of the GD, we uncover a phase-transition phenomenon for the case of Gaussian features: the classification error of GD is the same as that of the maximum-likelihood (ML) solution when $\kappa < \kappa_*$, and that of the max-margin (SVM) solution when $\kappa > \kappa_*$. Next, using the convex Gaussian min-max theorem (CGMT), we sharply characterize the performance of both the ML and SVM solutions. Combining these results, we obtain curves that explicitly characterize the test error of GD for varying values of κ . The numerical results validate the theoretical predictions and unveil “double-descent” phenomena that complement similar recent observations in linear regression settings.

Index Terms— Generalization error, Binary Classification, Overparameterization, Max-margin, Asymptotics

1. INTRODUCTION

Motivation. Modern learning architectures are chosen such that the number of training parameters overly exceeds the size of the training set. Despite their increased complexity, such *over-parametrized* architectures are known to generalize well in practice. In principle, this contradicts conventional statical wisdom of the so-called approximation-generalization tradeoff. The latter suggests a U-shaped curve for the test error as a function of the number of parameters, over which the error initially decreases, but increases after some point due to overfitting. In contrast, several authors uncover a peculiar W-shaped curve for the test error of neural networks as a function of the number of parameters. After the “classical” U-shaped curve, it is seen that a second approximation-generalization tradeoff (hence, a second U-shaped curve) appears for large enough number of parameters. The authors of [1, 2, 3] coin this the “double-descent” risk curve.

Recent efforts towards theoretically understanding the phenomenon of double-descent focus on linear regression with Gaussian features [4, 5, 6, 7]; also [8, 9, 10] for related efforts. These works investigate how the test error of gradient descent (GD) on square-loss depends on the overparameterization ratio $\kappa = p/n$, where p number of features are divided by the size n of the training set. On one hand, when $\kappa < 1$, GD iterations converge to the least-squares solution for which the generalization performance is well-known [4]. On the other hand, in the

overparameterized regime $\kappa > 1$, GD iterations converge to the min-norm solution for which the generalization performance is sharply characterized in [4, 6] using random matrix-theory (RMT). Using these sharp asymptotics, these papers identify regimes (in terms of model parameters such as SNR) for which a double-descent curve appears.

Contributions. This paper investigates the dependence of the classification error on the overparameterization ratio in binary linear classification with Gaussian features. In short, we obtain results that parallel previous studies for linear regression [4, 6]. In more detail, we propose studying gradient descent on logistic loss for two simple, yet popular, models: logistic model and gaussian mixtures (GM) model. Known results establish that GD iterations converge to either the support-vector machine (SVM) solution or the logistic maximum-likelihood (ML) solution, depending on whether the training data is separable or not. For the proposed learning model, we compute a phase-transition threshold κ_* and show that when problem dimensions are large, then data are separable if and only if $\kappa > \kappa_*$. Connecting the two, we redefine our task to that of studying the performance of ML and SVM. In contrast to linear regression, where the corresponding task can be accomplished using RMT, here we employ machinery based on Gaussian process inequalities. In particular, we obtain sharp asymptotics using the framework of the convex Gaussian min-max theorem (CGMT) [11, 12, 13]. Finally, we corroborate our theoretical findings with numerical simulations and build on the former to identify regimes where double-descent occurs.

Other related works. Our results on the performance of logistic ML and SVM fit in the rapidly growing recent literature on sharp asymptotics of (possibly non-smooth) convex optimization based estimators; [14, 13, 15] and many references therein. Most of these works study linear regression models, for which the CGMT framework has been shown to be powerful and applicable under several variations; see [13] and references therein. In contrast, our results hold for binary classification. For the derivations we utilize the machinery recently put forth in [16, 17, 18], which demonstrate that the CGMT framework can be also applied to classification problems. Closely related ideas were previously introduced in [19, 20]. Here, we introduce necessary adjustments to accommodate the needs of the specific data generation model and focus on classification error which has not been studied previously. There are several other works on sharp asymptotics of binary linear classification both for the logistic [21, 22, 18] and the GM model [23, 24]. While closely related, these works differ in terms of motivation, model specification, end-results, and proof techniques. The long version of this paper is available here [25].

2. LEARNING MODEL

We study supervised binary classification under the following data generative models: the logistic model and the Gaussian mixtures model.

Logistic model. Consider a planted model for the data in which

$$y = \begin{cases} +1, & \text{w.p. } \ell'(\mathbf{x}^T \boldsymbol{\eta}_0), \\ -1, & \text{w.p. } 1 - \ell'(\mathbf{x}^T \boldsymbol{\eta}_0), \end{cases} \quad \text{where } \mathbf{x} \in \mathbb{R}^d \text{ is a feature}$$

vector and $\boldsymbol{\eta}_0 \in \mathbb{R}^d$ is the unknown weight (or, regressor) vector and $\ell'(t) := (1 + e^{-t})^{-1}$. Throughout, we assume IID Gaussian feature vectors $\mathbf{x} \sim \mathcal{N}(0, \mathbf{I}_d)$. For compactness let $\text{Rad}(p)$ denote a symmetric Bernoulli distribution with probability p for the value $+1$ and probability $1 - p$ for the value -1 . We summarize the logistic model with Gaussian features:

$$y \sim \text{Rad}\left(\ell'(\mathbf{x}^T \boldsymbol{\eta}_0)\right), \quad \mathbf{x} \sim \mathcal{N}(0, \mathbf{I}_d). \quad (1)$$

Gaussian mixtures (GM) model. The second model assumes each data point belongs to one of two classes $\mathcal{C}_{\pm 1}$ with probabilities $\pi_{\pm 1}$. If it comes from class $\mathcal{C}_{\pm 1}$, then the feature vector $\mathbf{x} \in \mathbb{R}^d$ is an iid Gaussian vector with mean $\pm \boldsymbol{\eta}_0 \in \mathbb{R}^d$ and the response variable y takes the value of the class label ± 1 :

$$y = \pm 1 \Leftrightarrow \mathbf{x} \sim \mathcal{N}(\pm \boldsymbol{\eta}_0, \mathbf{I}_d). \quad (2)$$

2.1. Training data: Feature selection

During training, we assume access to n data points generated according to either (1) or (2). We allow for the possibility that only a subset $\mathcal{S} \subset [d]$ of the total number of features is known during training. Concretely, for each feature vector $\mathbf{x}_i$, $i \in [n]$, the learner only knows a sub-vector $\mathbf{w}_i := \mathbf{x}_i(\mathcal{S}) := \{\mathbf{x}_i(j), j \in \mathcal{S}\}$, for a chosen set $\mathcal{S} \subset [d]$. We denote the size of the *known* feature sub-vector as $p := |\mathcal{S}|$. Clearly, $1 \leq p \leq d$. Overall, the training set $\mathcal{T}(\mathcal{S})$ consists of n data pairs:

$$\begin{aligned} \mathcal{T}(\mathcal{S}) &:= \{(\mathbf{w}_i, y_i), i = 1, \dots, n\}, \\ \mathbf{w}_i &= \mathbf{x}_i(\mathcal{S}) \in \mathbb{R}^p \text{ where } (y_i, \mathbf{x}_i) \sim (1) \text{ or } (2). \end{aligned} \quad (3)$$

2.2. Classification rule

Having access to the training set $\mathcal{T}(\mathcal{S})$, the learner obtains an estimate $\boldsymbol{\beta}_* \in \mathbb{R}^p$ of the weight vector $\boldsymbol{\eta}_0 \in \mathbb{R}^d$. Then, for a newly generated sample $(\mathbf{x}, y)$ (and $\mathbf{w} := \mathbf{x}(\mathcal{S})$), she forms a *linear classifier* and decides a label $\hat{y}$ for the new sample as:

$$\hat{y} = \text{sign}(\mathbf{w}^T \boldsymbol{\beta}_*).$$

The estimate $\boldsymbol{\beta}_*$ is minimizing the empirical risk $\hat{\mathcal{R}}_{\text{emp}}(\boldsymbol{\beta}) := \frac{1}{n} \sum_{i=1}^n \ell(y_i \mathbf{w}_i^T \boldsymbol{\beta})$ for certain loss function $\ell : \mathbb{R} \rightarrow \mathbb{R}$. A traditional way to minimize $\hat{\mathcal{R}}_{\text{emp}}$ is by constructing gradient descent (GD) iterations $\boldsymbol{\beta}^{(k)}$, $k \geq 0$ via $\boldsymbol{\beta}^{(k+1)} = \boldsymbol{\beta}^{(k)} - \eta_k \nabla \hat{\mathcal{R}}_{\text{emp}}(\boldsymbol{\beta}^{(k)})$, for step-size $\eta_k > 0$ and arbitrary $\boldsymbol{\beta}^{(0)}$. We run GD until convergence and set $\boldsymbol{\beta}_* = \boldsymbol{\beta}^\infty := \lim_{k \rightarrow \infty} \boldsymbol{\beta}^{(k)}$. We focus on empirical logistic risk $\ell(t) := \log(1 + e^{-t})$.

Test error. For a new sample $(\mathbf{x}, y)$ we measure test error of $\hat{y}$ by the expected risk

$$\mathcal{R}(\boldsymbol{\beta}_*) := \mathbb{E}[\mathbb{1}(\hat{y} \neq y)]. \quad (4)$$

The expectation here is over the data generation model (1) or (2). In particular, here we consider the *excess risk*

$$\mathcal{R}_{\text{excess}}(\boldsymbol{\beta}_*) := \mathcal{R}(\boldsymbol{\beta}_*) - \mathcal{R}(\boldsymbol{\eta}_0),$$

where $\mathcal{R}(\boldsymbol{\eta}_0)$ is the risk of the best¹ linear classifier, i.e., $\mathcal{R}(\boldsymbol{\eta}_0) := \mathbb{E}[\mathbb{1}(y \neq \text{sign}(\mathbf{x}^T \boldsymbol{\eta}_0))]$.

The training error is $\mathcal{R}_{\text{train}}(\boldsymbol{\beta}_*) := \frac{1}{n} \sum_{i=1}^n \mathbb{1}(\hat{y} \neq y_i)$.

2.3. Convergence behavior of GD iterations

Recent literature studies the behavior of GD iterations for logistic loss. There are two regimes of interest: (i) when data is such that $\hat{\mathcal{R}}_{\text{emp}}(\boldsymbol{\beta})$ is strongly convex, then standard tools show that $\boldsymbol{\beta}$ converges to the unique bounded optimum of $\hat{\mathcal{R}}_{\text{emp}}(\boldsymbol{\beta})$; (ii) when data is separable, then the normalized iterations $\boldsymbol{\beta}^{(k)} / \|\boldsymbol{\beta}^{(k)}\|_2$ converge to the max-margin solution [26, 27]. These properties guide our approach towards studying the test error of GD for logistic loss. Instead of the latter, we study the performance of the max-margin classifier and of the minimum of the empirical logistic risk. We formalize these ideas next.

Separable data. The training set is separable iff there exists a linear classifier achieving zero training error, i.e., $\exists \boldsymbol{\beta} : y_i \mathbf{w}_i^T \boldsymbol{\beta} \geq 1, \forall i \in [n]$. When data is separable, for $k \rightarrow \infty$ it holds $\left\| \frac{\boldsymbol{\beta}^{(k)}}{\|\boldsymbol{\beta}^{(k)}\|_2} - \frac{\hat{\boldsymbol{\beta}}}{\|\hat{\boldsymbol{\beta}}\|_2} \right\|_2 \rightarrow 0$, where

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \|\boldsymbol{\beta}\|_2 \quad \text{subject to } y_i \mathbf{w}_i^T \boldsymbol{\beta} \geq 1, \quad (5)$$

i.e., the solution to **hard-margin SVM**.

Non-separable data. When the separability condition does *not* hold, then $\hat{\mathcal{R}}_{\text{emp}}(\boldsymbol{\beta})$ is coercive. Thus, its sub-level sets are closed and GD iterations will converge [26] to the minimizer $\hat{\boldsymbol{\beta}}$ of the empirical loss:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \sum_{i=1}^n \ell(y_i \mathbf{w}_i^T \boldsymbol{\beta}), \quad (6)$$

where, recall that $\ell(t) = \log(1 + \exp(-t))$. For the data model in (1), $\hat{\boldsymbol{\beta}}$ is the **maximum-likelihood (ML)** estimator.

3. SHARP ASYMPTOTICS

3.1. Setting

Recall the following notation:

- d : dimension of the ambient space,
- n : training sample size,
- p : number of parameters used for training (see (3)).

We study a setting in which d and n are fixed and p varies from 1 to d . Our asymptotic results hold in a *linear asymptotic regime* where $n, d, p \rightarrow +\infty$ such that

$$d/n \rightarrow \zeta > 0 \quad \text{and} \quad p/n \rightarrow \kappa \in (0, \zeta]. \quad (7)$$

We fix $\zeta \in [1, +\infty]$ and derive asymptotic predictions for the test error as a function of $\kappa \in (0, \zeta]$, which determines the overparameterization ratio in our model. To quantify the effect of κ on the test error, we decompose the feature vector $\mathbf{x}_i \in \mathbb{R}^d$ to its *known* part $\mathbf{w}_i \in \mathbb{R}^p$ and to its unknown part $\mathbf{z}_i$: $\mathbf{x}_i := [\mathbf{w}_i^T, \mathbf{z}_i^T]^T$. Similarly, we let $\boldsymbol{\beta}_0 \in \mathbb{R}^p$ (resp., $\boldsymbol{\gamma}_0 \in \mathbb{R}^{d-p}$) denote the vector of weight coefficients corresponding to the known (resp., unknown) features such that $\boldsymbol{\eta}_0 := [\boldsymbol{\beta}_0^T, \boldsymbol{\gamma}_0^T]^T$.

¹It assumes knowledge of the entire feature vector $\mathbf{x}_i$ and of $\boldsymbol{\eta}_0$.

In this notation, we study a sequence of problems of increasing dimensions as in (7) that further satisfy:

$$\begin{aligned} \|\eta_0\|_{2 \rightarrow r} &\rightarrow r, \\ \|\beta_0\|_{2 \rightarrow s} &:= s(\kappa) \quad \text{and} \quad \|\gamma_0\|_{2 \rightarrow \sigma} := \sigma(\kappa). \end{aligned} \quad (8)$$

The parameters s and σ can be thought of as the useful signal strength and the noise strength, respectively. Our notation specifies that $s(\kappa)$ (hence also, $\sigma(\kappa) = \sqrt{r^2 - s^2(\kappa)}$) is a function of κ . We are interested in functions $s(\kappa)$ that are increasing in κ such that the signal strength increases as more features enter into the training model; Sec. 4.1 for explicit parameterizations.

Supplementary results. Due to space constraints, we only present results for the logistic model. Additional results and all proofs can be found in [25].

Notation. We reserve the following notation

$$H, G, Z \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), Y \sim \text{Rad}(\ell'(sG + \sigma Z)), \quad (9)$$

for s and σ as defined in (8). All expectations and probabilities are with respect to the randomness of H, G, Z . Also, let $(x)_- = \min\{x, 0\}$. For a sequence of random variables $\mathcal{X}_{n,p,d}$ that converges in probability to a constant c in the limit of (7), we simply write $\mathcal{X}_{n,p,d} \xrightarrow{P} c$. Finally, we denote the proximal operator of the logistic-loss as follows, $\text{prox}_\ell(x; \lambda) := \arg \min_v \frac{1}{2\lambda}(x - v)^2 + \ell(v)$.

3.2. Regimes of learning

As discussed in Section 2, the behavior of GD changes depending on whether the training data is separable or not. Under the Gaussian feature model, we establish a sharp phase-transition characterizing the separability of the training data.²

Proposition 3.1 (Phase transition – Logistic model). *Consider the logistic model with Gaussian features in (1). Let $\mathcal{E}_{\text{sep}} := \{\exists \beta : y_i \mathbf{w}_i^T \beta \geq 1, \forall i \in [n]\}$ be the event under which the training data are separable. Let $s = s(\kappa)$ in (8) be an increasing function of κ and, recalling (9), define $g(\kappa) := \min_{t \in \mathbb{R}} \mathbb{E}(H + tGY)^2_-$. Let κ_* be the unique positive solution to the equation $g(\kappa) = \kappa$. Then, the following holds regarding $\mathcal{E}_{\text{sep}}$:*

$$\begin{aligned} \kappa > \kappa_* &\Rightarrow \lim_{n,p,d \rightarrow \infty} \Pr(\mathcal{E}_{\text{sep}}) = 1, \\ \kappa < \kappa_* &\Rightarrow \lim_{n,p,d \rightarrow \infty} \Pr(\mathcal{E}_{\text{sep}}) = 0. \end{aligned}$$

Put in words: the training data is separable iff $\kappa > \kappa_*$. When this is the case, then the training error $\mathcal{R}_{\text{train}}$ can be driven to zero and we are in the *interpolating regime*. In contrast, the training error is non-vanishing for smaller values of κ .

3.3. Asymptotics

Propositions 3.2 and 3.3 characterize the performance of the two optimization-based classifiers in (10) and (5), respectively. When combined with the results of Section 2.3, they characterize the statistical performance of converging points of GD.

²In slightly different setting, similar phase-transition results have been obtained recently under both the logistic [21] and the GM [16] models. Specifically, compared to the result in [21], Proposition 3.1 further involves the following: (i) an extension to the noisy setting; (ii) a proof that the equation $g(\kappa) = \kappa$ has a unique solution.

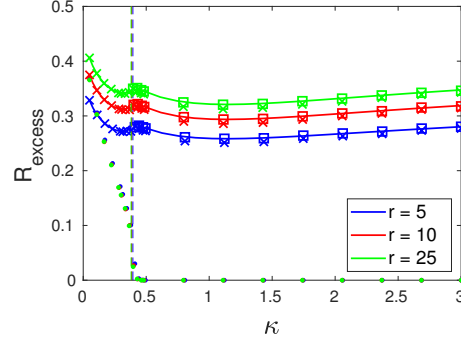


Fig. 1: Excess risk (aka test error) of GD iterates on logistic loss as a function of the overparameterization ratio κ for the logistic data model in (1) with polynomial feature selection as in (12). We show: (i) Monte Carlo results for excess risk ('x') and training error ('•') of GD; (ii) Monte Carlo results for excess risk of SVM ('□'); and, (iii) theoretical predictions for excess risk (solid lines). Different colors correspond to different values of total signal strength $r = 5, 10$ and 25 . The dashed lines (almost indistinguishable due to scaling of the figure) indicate the values of κ_* separating the underparameterized from the overparameterized regimes.

The performance of logistic loss under the logistic model was recently studied in [22, 18, 17]. Here, we follow the notation in [17, Thm. III.1] and appropriately modify the result to fit the data model of Section 2. Also, different to prior work, our predictions hold for the classification error.

Proposition 3.2 (Performance of ML). *Fix $\kappa < \kappa_*$. Let (μ, α, λ) be the unique solution to the following system of three nonlinear equations in three unknowns.*

$$\begin{aligned} \mu &= \mathbb{E}[YG \cdot \text{prox}_\ell(\alpha H + \mu YG; \lambda)], \\ \alpha^2 \kappa &= \lambda^2 \mathbb{E}[(\ell'(\text{prox}_\ell(\alpha H + \mu YG; \lambda)))^2], \\ \kappa &= \lambda \mathbb{E}\left[\frac{\ell''(\text{prox}_\ell(\alpha H + \mu YG; \lambda))}{1 + \lambda \ell'''(\text{prox}_\ell(\alpha H + \mu YG; \lambda))}\right]. \end{aligned}$$

Then, $\mathcal{R}(\hat{\beta}) \xrightarrow{P} \mathbb{P}(\mu GY + \alpha H < 0)$.

We characterize the asymptotic performance of SVM under the logistic model. Corresponding results for the GM model are derived in [16].

Proposition 3.3 (Performance of SVM). *Let $\kappa > \kappa_*$ and define $\eta(q, \rho) := \mathbb{E}(\rho GY + \sqrt{1 - \rho^2} H - \frac{1}{q})^2_- - (1 - \rho^2)\kappa$. Let q_* be the unique solution of the equation $\min_{-1 \leq \rho \leq 1} \eta(q, \rho) = 0$. Further let ρ_* be the unique minimum of $\eta(q_*, \rho)$ for $\rho \in [-1, 1]$. Then, $\mathcal{R}(\hat{\beta}) \xrightarrow{P} \mathbb{P}(\rho_* GY + \sqrt{1 - \rho_*^2} H < 0)$.*

4. NUMERICAL RESULTS

4.1. Feature selection models

Recall that the number of features known at training is determined by κ . Specifically, κ enters the formulae predicting the classification performance via the signal strength $s = s(\kappa)$. In this section, we specify two explicit models for feature selection and their corresponding functions $s(\kappa)$. Similar models are considered in [28, 4, 6], albeit in a linear regression setting.

Linear model. We start with a uniform feature selection model characterized by the following parametrization:

$$s^2 = s^2(\kappa) = r^2 \cdot (\kappa/\zeta), \quad \kappa \in (0, \zeta], \quad (11)$$

for fixed r^2 and $\zeta > 0$. This models a setting where all coefficients of the regressor η_0 have equal contribution. Hence, the signal strength $s^2 = \|\beta_0\|_2^2$ increases linearly with the number p of features used in training.

Polynomial model. In the linear model, adding more features during training results in a linear increase of the signal strength. In contrast, our second model assumes diminishing returns:

$$s^2 = s^2(\kappa) = r^2 \cdot (1 - (1 + \kappa)^{-\gamma}), \quad \kappa > 0, \quad (12)$$

for some $\gamma \geq 1$. As κ increases, so does the signal strength s , but the increase is less significant for larger values of κ at a rate specified by γ .

4.2. Risk curves

Figure 1 assumes the polynomial feature model (12) for $\gamma = 2$ and three values of r . The crosses (' $\times$ ') are simulation results obtained by running GD on synthetic data generated according to (1) and (12). Specifically, the depicted values are averages calculated over 500 Monte Carlo realizations for $p = 150$. For each realization, we ran GD on logistic loss (see Sec. 2.2) with a fixed step size until convergence and recorded the resulting risk. Similarly, the squares (' $\square$ ') are simulation results obtained by solving SVM (5) over the same number of different realizations and averaging over the recorded performance. As expected by [26] (also Sec. 2.3), the performance of GD matches that of SVM when data is separable. Also, as predicted by Proposition 3.1, the data is separable with high-probability when $\kappa > \kappa_*$ (the threshold value κ_* is depicted with dashed vertical lines). This is verified by noticing the dotted (' $\bullet$ ') scatter points, which record (averages of) the training error. Recall from Section 2.2 that the training error is zero if and only if the data are separable. Finally, the solid lines depict the theoretical predictions of Proposition 3.2 ($\kappa < \kappa_*$) and Proposition 3.3 ($\kappa > \kappa_*$). The results of Figure 1 validate the accuracy of our predictions: our asymptotic formulae predict the classification performance of GD on logistic loss for all values of κ .

Figures 3 and 2 depict the excess risk of GD as a function of κ for the linear and the polynomial model, respectively. Compared to Figure 1, we only show the theoretical predictions. Note that the linear model is determined by parameters (ζ, r) and the polynomial model by (γ, r) . Once these values are fixed, we compute the threshold value κ_* . Then, we apply Proposition 3.2 ($\kappa < \kappa_*$) and Proposition 3.3 ($\kappa > \kappa_*$).

4.3. Discussion on double-descent

Here, we discuss the double-descent phenomenon that appears in Figures 1 and 2. First, focus on the underparameterized regime $\kappa < \kappa_*$ (cf. area on the left of the vertical dashed lines). Here, the number n of training examples is large compared to the size p of the unknown weight vector β_0 , something that favors learning β_0 . However, since only a (small) fraction p/d of the total number d of features is used for training, the observations are rather noisy. This tradeoff manifests itself with a U-shaped curve for $\kappa < \kappa_*$. Such a tradeoff is also present in the overparameterized regime $\kappa > \kappa_*$ (cf. area on the right of the vertical dashed lines) leading to a second U-shaped curve. The larger the value of κ the more accurate the model, but also

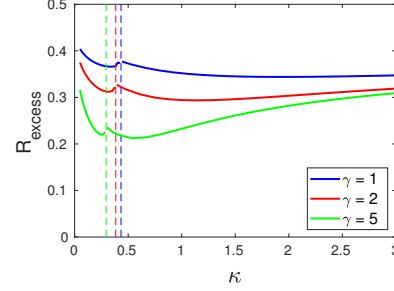


Fig. 2: Plots of the excess risk (right) as a function of κ for the polynomial feature selection model (cf. (12)) with $r = 10$ and three values of $\gamma = 1, 2$ and 5 .

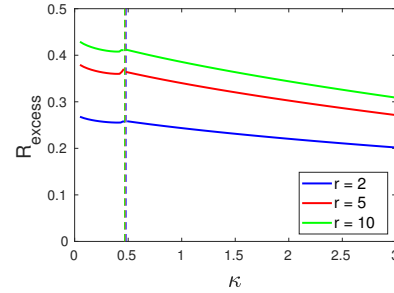


Fig. 3: Plots of the excess risk (right) as a function of κ for the linear feature selection model (cf. (11)) with $\zeta = 3$ and three values of $r = 2, 5$ and 10 .

larger the number of unknown parameters to be estimated. As seen in Figure 2, the U-shape is more pronounced for larger values of γ . Overall, the risk curve has two local minima corresponding to the two regimes of learning. Interestingly, the global minimum κ_{opt} appears in the overparameterized regime for all values of r and γ depicted here. The value of $\kappa_{\text{opt}} > \kappa_*$ determines the optimal number of features that need to be selected during training to minimize the classification error. Note that for κ_{opt} the training error is zero, yet the classification performance is best. This observation is consistent with existing (non-sharp) results on the favorable generalization properties of max-margin classifiers such as SVM, e.g., [29].

The risk curves for the linear feature selection model in Figure 3 are somewhat different compared to the polynomial model. In the underparameterized regime, we observe a (very shallow) U-shaped curve, but the risk is monotonically decreasing for $\kappa > \kappa_*$. For the values of signal strength r considered here the global minimum is always attained at $\kappa = \zeta = 3$, i.e., when all features are used for training.

5. FUTURE WORK

Due to space constraints, Sections 3 and 4 present results only for the logistic model. However, note that the proposed learning model of Sec. 2.1 as well as the connection between GD iterates and the ML and SVM solutions further apply to the GM model. We include these extensions in [25]. Here, we study the generalization error of GD on logistic loss. In future work, we plan on extending our study to GD on square loss, which is also commonly used in practice. Extensions of the results to multi-class settings are also of interest.

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